

Week 7: Grounding audit — is T5-small extracting from the prompt?

Auto-generated 2026-05-15 18:25 from outputs/week07_grounding/. Do not edit by hand. Re-run `python -m src.reporting.build_report --week week07` to refresh.

1. Objective

W3/W5/W6 established that **reranked retrieval roughly doubles every W3 generation metric** (Token-F1 +0.171, ROUGE-L +0.174, both 95 % CIs strictly above zero), that this is **not a surface-form artefact** (BERTScore proxy Δ +0.173), and that the **decode budget is not the bottleneck** (mnt=64 $\rightarrow$ 128 closure run did not move metrics or shrink the regression bucket). One question that remained unanswered: *what is T5-small actually doing on this prompt format?* Specifically:

Is the model **extracting** content from the retrieved passages, or **generating** from its parametric memory and happening to overlap?

This is the *grounding* question. It bounds how much of the W3 gain should be attributed to retrieval (passage content makes it into the prompt) vs to the generator (content is produced de novo). W7 answers it with two cheap deterministic metrics computed on the **already-on-disk W3 predictions** — no new generation, no model load.

2. Setup

Inputs: the same two paired prediction files used by every W3/W5/W6 analysis script — `outputs/week03_generation_bm25_full/predictions.jsonl` and `outputs/week03_generation_reranked_full/predictions.jsonl`. The key property is that each row already contains the *exact* top-K passage texts that the generator saw at prompt time, alongside the prediction. So grounding can be scored without re-running anything.

Two metrics, both pure-Python regex-tokenised CPU functions (`src/evaluation/grounding.py`, ~150 lines, no model load, no NLTK or sklearn dependency):

- **Lexical content-token grounding.** Fraction of unique non-stopword tokens in the prediction (lowercased) that appear anywhere in the union of passage texts. A local 85-word stopwords list keeps the metric self-contained.
- **3-gram grounding.** Fraction of the prediction's contiguous 3-grams (case-folded, stopwords retained, since n-gram order is the signal) that appear as a contiguous span in at least one passage. Catches phrase-level extractiveness the lexical metric is blind to.

Edge cases (documented, stable, *reported separately*): predictions with no content tokens score lexical 1.0 vacuously; predictions shorter than n tokens score n-gram 1.0 vacuously. The audit reports the count of such predictions per arm so this convention cannot silently inflate the headline number.

NLI-based grounding (entailment of prediction by passages) is the canonical semantic faithfulness metric and is the obvious extension; it is **deliberately not in this commit** (see §7 *Next*).

3. Lexical content-token grounding

Per-query metric: fraction of the prediction's unique non-stopword tokens that appear anywhere in the prompt's passage union. 1.0 means every distinct content word the model emitted is present in one of the top-K passages it was conditioned on.

Metric	BM25 (mean)	Reranked (mean)	Δ (rerank – BM25)	95 % CI	p (two- sided)
Lexical (content- token)	0.9972	0.9977	+0.0005	[-0.0003, +0.0014]	0.238

Per-query split:

Outcome	Rate
Rerank strictly more grounded	2.3 %
Tie	95.8 %
BM25 strictly more grounded	1.9 %

Headline read. Both arms sit at the lexical ceiling — almost every content word the model emits is already in the prompt. T5-small on the question: ... context: ... shape is effectively performing **extractive QA**, not generative QA. The Δ between arms is tiny (+0.0005); the rerank advantage on lexical grounding alone is near-zero because there is essentially no slack for retrieval to take advantage of at the *word* level.

4. 3-gram grounding

Per-query metric: fraction of the prediction's contiguous 3-grams that match a contiguous span in any single passage. Tighter than the lexical metric — catches "the model strung together the right words in the wrong order" cases that the bag-of-content-tokens score misses.

Metric	BM25 (mean)	Reranked (mean)	Δ (rerank – BM25)	95 % CI	p (two- sided)
3- gram	0.9873	0.9905	+0.0032	[+0.0015, +0.0050]	< 0.001

Per-query split:

Outcome	Rate
Rerank strictly more grounded	10.1 %
Tie	83.2 %
BM25 strictly more grounded	6.7 %

Both arms remain near the ceiling at the phrase level too. The Δ is small but the CI strictly excludes zero: reranking does push 3-gram grounding up — slightly — because better-ranked passages contain contiguous phrasings closer to the gold answer.

5. Edge cases

Predictions that fall into the documented "vacuous $\rightarrow$ 1.0" path of each metric (see `src/evaluation/grounding.py` docstrings):

Arm	Lex vacuous (no content tokens)	3-gram vacuous (<3 tokens)
BM25	15	693
Rerank	7	735

The 3-gram vacuous counts (~10 % of 6,980 shared qids per arm) are exactly the `truncation_short / extractive_passage_bias` pattern surfaced by the W6 regression taxonomy — predictions of 1–2 tokens, often a passage title. They inflate the n-gram headline by a few tenths of a percentage point *identically on both arms*, so the paired Δ is unaffected. The lexical vacuous counts are negligible (<0.3 % of either arm).

6. Limitations

- **Extractiveness, not faithfulness.** The two metrics measure whether the prediction *literally derives from* the prompt passages. A heavy paraphrase that is perfectly faithful semantically will score low; a verbatim copy from a *distractor* passage will score high. The right tool for the semantic faithfulness question is a small NLI cross-encoder over (passages $\rightarrow$ prediction); it is listed in §7 *Next*.
- **Top-K passages only.** Grounding is measured against the top-3 passages T5-small actually saw at prompt time, not against the full retrieved top-100 or the broader corpus. This is the relevant notion of grounding for *this* generator: the question is "did the model use the prompt", not "is the answer findable somewhere in MS MARCO".
- **Word- and phrase-level signal only.** Both metrics are bag-of- unigrams / bag-of-3-grams. They do not see word order beyond $n=3$, nor any morphology / synonymy / tense variation. A future NLI pass would close all three gaps.
- **The ceiling effect is the headline.** Because both arms sit at ~99 % lexical grounding, the W7 rerank-vs-BM25 Δ for grounding is essentially zero. This is a *finding*, not a limitation, but it does mean the audit's primary value here is the *level*, not the Δ : it bounds how much of the W3 surface-form rerank gain is attributable to the generator vs to retrieval. The answer is "almost none to the generator at this prompt format".

7. Next

- **NLI-based grounding.** Score entailment (passages $\rightarrow$ prediction) on a 3 000-pair subsample with a small CPU cross-encoder (cross-encoder/nli-deberta-v3-small). Mirrors the W6 BERTScore- proxy pattern: cheap, deliberately not citation-grade, and tests whether the *semantic* grounding Δ matches the *lexical* one. The audit script already exposes `--nli-n-pairs` as a placeholder argument; the metric is the only thing missing.
- **Bucket / query-type breakdown.** Join the per-query grounding scores with the W6 `per_query_metrics.jsonl` and regression bucket labels. Specifically, ask: are the W3 regression queries (rerank brings relevant passage to top-3, generator gets *worse* F1 than under BM25) characterised by *passage-external* predictions on the rerank arm? If yes, the W6 truncation taxonomy's `truncation_midword` rule is identifying low-grounding outputs, not budget-cut outputs — a cleaner mechanism for the W6 closure finding.
- **Span-level citation linking.** For each prediction, attribute its content tokens to the specific source passage. Cheap (~50 lines) but explicitly out of scope for this commit per the W7 scope.

8. Reproduce

```
# The audit runs on the existing W3 paired-prediction files. No
# new
# generation, no model load; CPU, ~2 minutes end-to-end (~2 s
# scoring
# + the paired bootstrap on two metrics).
python scripts/grounding_audit.py \
    --bm25-dir outputs/week03_generation_bm25_full \
    --reranked-dir outputs/week03_generation_reranked_full \
    --output-dir outputs/week07_grounding
```

Outputs (gitignored):

- `outputs/week07_grounding/per_query_grounding.jsonl` — per-qid scores for both arms and both metrics, ready for join with the W6 bucket / query-type tables.
- `outputs/week07_grounding/summary.json` — means, paired-bootstrap CIs, win/tie/loss, edge-case diagnostics, full provenance.